PORTFOLIO · 2026

Shangze Kong

Robotics · Embodied Intelligence
Multimodal Perception
Industrial Automation

BEng Mechanical Engineering
University College London · 2024 – 2027

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ABOUT

Engineer at the intersection of robotics, multimodal perception, calibration, and control.

Mechanical Engineering undergraduate at UCL with a First-class average and a research record spanning industrial robotics, autonomous navigation, and vision-language agents. I approach robotics through the full perception–calibration–control pipeline, studying how multimodal visual and sensor evidence can be converted into reliable state estimates, motion plans, and closed-loop actions. My experience includes multimodal perception research, external virtual TCP design, vision-based calibration and AGV closed-loop correction in a Fudan robotics project, as well as navigation and control work in a Cambridge project.

73/100

First-class average
UCL Year 1

2 papers

1 published
1 under review

ACM MM

2026

CCF-A submission
under review

3 lab experiences

Research positions
Fudan · Cambridge · UCL

EDUCATION

University College London

BEng Mechanical Engineering · 2024 – 2027

Year 1 average: 73 / 100 (UCL First-Class band)

Year 2 average: 71 / 100 (UCL First-Class band)

RESEARCH INTEREST

Embodied Intelligence, Visual perception for robotic localisation, hand–eye calibration, multi-frame fusion, and closed-loop execution — and how these inform motion planning and control under uncertainty.

TOOLKIT

ROS / MoveIt · OpenCV · PyTorch · Transformers · Real-ESRGAN · Grounding DINO · SAM 2 · Depth Anything V2 · MATLAB / Simulink · SolidWorks · Python · C++ · LaTeX · Linux

Mapping the perception ceiling of frozen multimodal models

ACM MM 2026 · CCF-A · First author, under review · Solo lead · Oct 2025 – Apr 2026

BACKGROUND

While training perception modules into VLMs delivers gains, the results scale unpredictably with data and architecture, and remain checkpoint-locked. I asked a different question: what is the upper bound a frozen model can reach if given ideal access to visual evidence?

FRAMEWORK

A tool-assisted oracle ceiling: per-sample best-tool selection that cleanly separates information-access failure from capacity failure in multimodal reasoning.

+27.8%

Oracle relative gain over Qwen2-VL-7B baseline (0.661 → 0.845)

MAIN RESULTS

Method	Train	Tools	CUB	Flickr	GQA	POPE	VSR	Avg.	Δ_{BL}
Qwen2-VL-7B									
Baseline	✗	0	.558±.043	.634±.032	.608±.042	.867±.033	.640±.043	.661±.018	–
VPT (GD) [23] †	522k	1	**.930**±.022	.720±.030	.590±.042	.883±.031	.730±.040	.771±.016	+16.6%
VPT (Full) [23] †	522k	2	.921±.024	.680±.031	.623±.042	.903±.029	**.788**±.036	.783±.016	+18.5%
PercepAgent (ours)	✗	4	.800±.035	**.740**±.029	**.673**±.041	**.940**±.023	.710±.040	.773±.016	+16.9%
Oracle ‡	✗	4	.919±.024	.790±.027	.753±.037	.945±.022	.817±.034	.845±.014	–
Qwen3-VL-8B									
Baseline	✗	0	.720±.039	.814±.026	.520±.043	.900±.029	.687±.041	.728±.017	–
PercepAgent (ours)	✗	4	**.850**±.031	**.820**±.026	**.710**±.039	**.950**±.021	**.700**±.041	**.806**±.015	+10.7%
Oracle ‡	✗	4	.940±.021	.850±.024	.743±.038	.940±.023	.852±.032	.865±.013	–
Single-tool variants (Qwen2-VL-7B)									
+ SR only	✗	1	.678±.041	**.694**±.031	.635±.042	.905±.028	.672±.042	.717±.017	+8.5%
+ GD only	✗	1	.597±.043	.565±.033	.618±.042	**.919**±.026	.648±.043	.669±.018	+1.2%
+ DE only	✗	1	.697±.040	.658±.032	.624±.042	.913±.027	**.691**±.041	.717±.017	+8.3%
+ Seg only	✗	1	**.756**±.037	.649±.032	**.651**±.041	.916±.027	.680±.042	.731±.017	+10.6%
Tool-selection strategy (Qwen2-VL-7B)									
Random-tool	✗	4	.740±.038	.698±.031	.560±.043	.900±.029	.676±.042	.715±.017	+8.2%
All-tools	✗	4	.760±.037	**.750**±.029	.600±.042	.910±.028	.592±.044	.722±.017	+9.2%
ReActAgent [22]	✗	4	.690±.040	.591±.032	.610±.041	.840±.035	.580±.044	.662±.018	+0.1%

Oracle (0.845) exceeds VPT fine-tuning on 522k samples (0.783) — without any training.

PercepAgent: a minimal training-free agent that reaches the ceiling

The agent is the instrument, not the conclusion — built deliberately simple to test whether the ceiling is reachable in practice, not just measurable in theory.

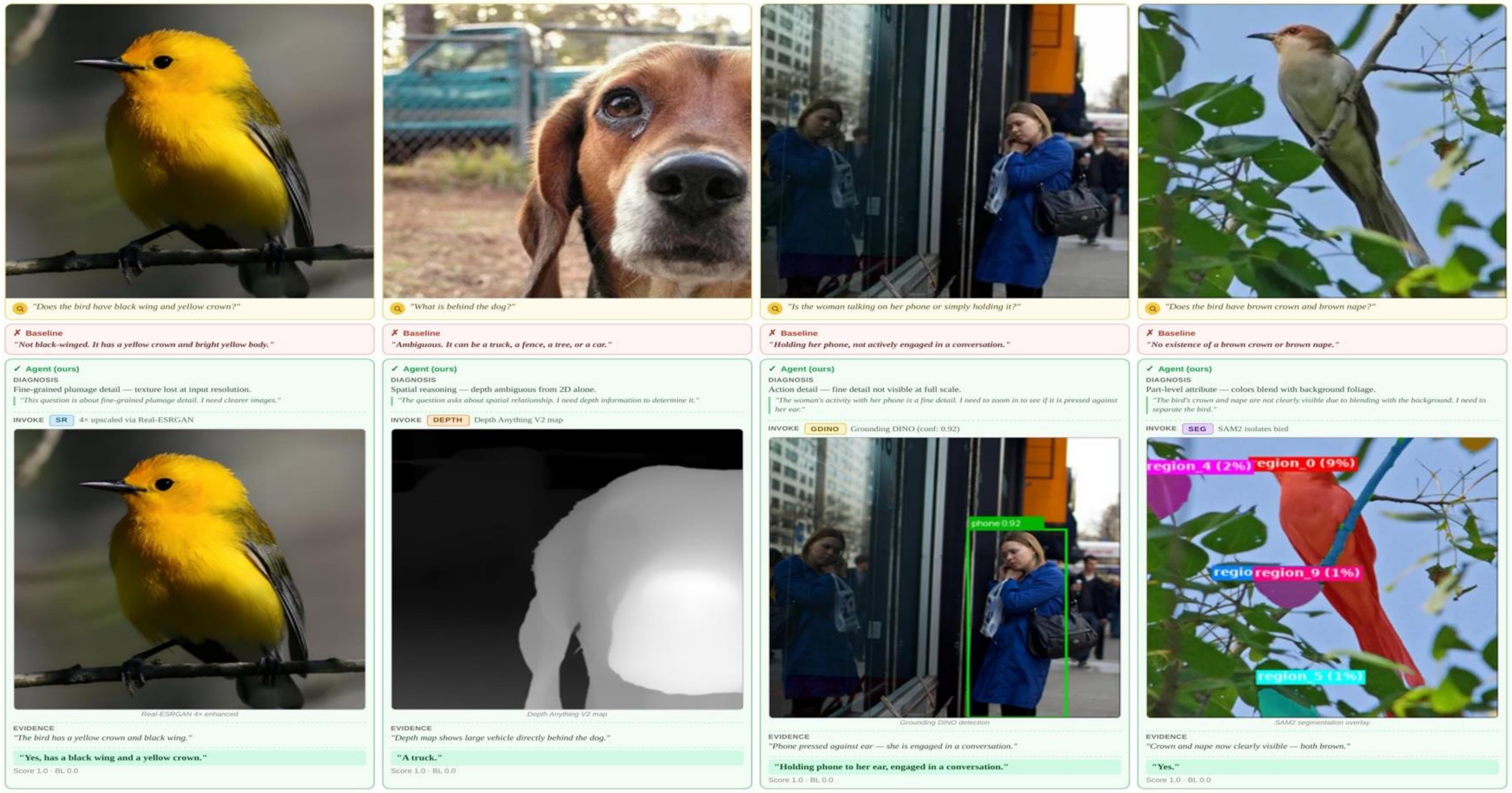


Figure 1. VLMs fail on visually grounded tasks not due to reasoning, but due to perceptual information gaps. Our training-free agent diagnoses the failure type (with explicit chain-of-thought reasoning), then invokes the appropriate perception tool — super-resolution (SR), depth estimation (Depth), grounding detection (GDino), or segmentation (Seg) — to recover the missing visual evidence, consistently correcting baseline errors across four distinct perception gap categories.

Figure: training-free agent diagnoses the perception gap and invokes SR / Depth / Grounding DINO / Seg to recover missing visual evidence.

RECOVERY

61%

of oracle headroom recovered
at zero training cost

DESIGN

Evidence-gap diagnosis

Asks what visual evidence is missing — not how confident the model is.

Baseline-first safety gate

Wrong tool calls are contained, not committed; performance never falls below the frozen baseline.

Vision-guided sub-millimetre robotic alignment for sewing-template handling

Fudan University · Institute of AI & Robotics · Full-time research assistant · Advisor: Prof. Qi Lizhe · Jul – Sep 2025

WHY THIS WORK MATTERS

Sewing automation is mature on the machine side — but template loading still relies on manual labour. Glass-fibre templates abrade and irritate the lungs and skin of operators, and ageing labour pools are pushing the industry toward humanoid template-handling robots.

Mobile manipulator platforms developed at the Fudan Institute of AI & Robotics.

PROBLEM

Vendor-locked robot-arm TCP; AGV docking error too large for direct teaching. A non-invasive precision recovery scheme was required.

METHOD

External virtual TCP via a three-marker orthogonal target; pixel-to-end-effector mapping; AGV closed-loop pose correction integrated into one pipeline.

OUTCOME

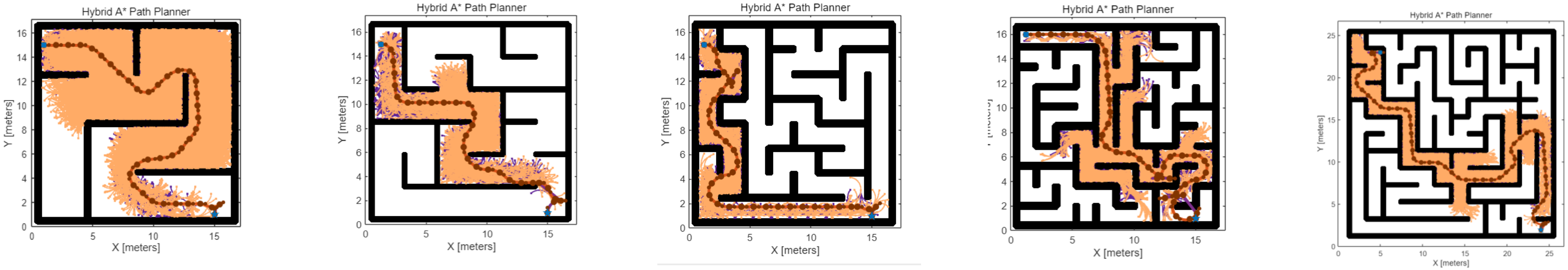
Achieved $\pm$0.5 mm grasping accuracy and <0.5° rotation error without firmware modification; validated end-to-end and exhibited at CISMA 2025.

03

ROBUST NAVIGATION · TEAM LEAD

Multi-sensor fusion & cooperative perception for autonomous robots in complex maze environments

CONFMLA 2025 · First author, published · University of Cambridge · Advisor: Prof. Fumiya Iida · Jun – Sep 2025



Hybrid A path planner solving 5-, 7-, 9-, 11-, and 13-branch mazes (left → right) — 100% success across complexity at the cost of compute.*

CONTRIBUTION

- Led 6-person team to build a unified evaluation harness on binary occupancy grids.
- Proposed a deadlock-diagnosis framework: progress monitor, state-hash, hit/leave replay.
- Lightweight perception–control coupling with hysteresis and no-return zones for escape.
- Validated across 5 branching factors × 10 trials with Mann–Whitney U significance.
- Publication: Shangze Kong. Multi-Sensor Fusion Methods and Cooperative Perception for Autonomous Robots in Complex Maze Environments. CONFMLA 2025

ALGORITHM COMPARISON · SUCCESS RATE

Algorithm	5-branch	13-branch
A* / Hybrid A* *Reliable, slow*	100%	100%
Bug2 *Fast, deadlocks*	100%	10%
Braitenberg Vehicle *Reactive only*	90%	<5%

Thank you for reading.

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